

# A Size-Invariant Deep Reinforcement Learning Policy for the Interval Job Shop Scheduling Problem

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## Abstract

Task durations on a shop floor are seldom known exactly when scheduling decisions must be made. The interval job shop scheduling problem (IJSP) models each processing time as a closed interval; published solvers search each instance separately with metaheuristics. This paper studies the complementary regime: a constructive policy, trained once by deep reinforcement learning, that dispatches operations one at a time and returns a schedule in milliseconds. It is size-invariant by construction: dimensionless features and a permutation-invariant encoder give policy and value heads a parameter count independent of the numbers of jobs and machines. Trained with proximal policy optimization on four **20×15** instances of the interval Taillard benchmark, it reaches **13.8%** mean relative error against crisp lower bounds on unseen instances of that size, against  $\approx$ **46%** for the best dispatching rule there, and beats that rule zero-shot on every tested instance of all seven size classes; the same weights transfer to a second benchmark of classical instances. Against an evolved dispatching rule the ordering depends on the inference budget: pass for pass the rule wins, and at **1024** samples each the policy wins 46 of the 70. Three ablations are negative: self-attention does not improve on pooling; multi-size training does not overtake a single-size specialist at three times its cost; and the interval-width features can be removed at no cost, as the closed form of the worst-case makespan predicts. Training on crisp midpoints does cost **0.63** points: what the interval model buys lies in the data rather than the features.

**Keywords:** job shop scheduling, interval uncertainty, deep reinforcement learning, neural combinatorial optimization, size invariance, dispatching

# 1 Introduction

A production schedule is committed before the work it plans has been done, and the durations it assumes are estimates: tool wear, operator variability and material differences all move them. The job shop scheduling problem, NP-hard (Garey, Johnson, & Sethi, 1976) and industrially ubiquitous, is nonetheless posed in its textbook form as if every processing time were known exactly. Of the formalisms proposed for this uncertainty, stochastic and fuzzy models (González-Rodríguez, Vela, & Puente, 2010) demand distributions or membership functions from the decision maker; the *interval* model asks only for bounds. In the resulting interval JSP (IJSP), each duration is a closed interval, schedule evaluation propagates those intervals, and the makespan is itself an interval, compared through a ranking criterion (Díaz, Palacios, González-Rodríguez, & Vela, 2023a; Lei, 2011).

Makespan minimization in the IJSP is currently the territory of per-instance metaheuristic search: elitist artificial bee colony algorithms (Díaz et al., 2023a; Díaz, Palacios, González-Rodríguez, & Vela, 2023b) reach mean relative errors (RE, measured against crisp Taillard lower bounds) between roughly 6% and 16% depending on the size class. The price is structural: every new instance restarts the search from scratch, for seconds to minutes per run on dedicated hardware.

The deterministic scheduling literature has developed a complementary paradigm, *learning to dispatch*: a deep reinforcement learning (DRL) agent builds the schedule constructively, choosing one eligible operation at a time (Park, Chun, Kim, Kim, & Park, 2021; Tassel, Gebser, & Schekotihin, 2021; Zhang et al., 2020). The training cost is paid once; afterwards the policy prices any instance in milliseconds, which makes it attractive both as a standalone solver under tight time budgets and as an initializer for search. Yet this paradigm has not crossed over to interval durations: a recent survey of learning-based job shop scheduling (Xu, Mei, Zhang, & Zhang, 2025) lists no method addressing interval-valued processing times.

Two obstacles stand between standard JSP-DRL designs and the IJSP. The first is representational: uncertainty must be visible to the agent, which should treat an operation lasting  $[10, 11]$  differently from one lasting  $[5, 16]$  even though both share the same midpoint. The second is architectural: common designs tie network dimensions to the instance size, so a policy trained on one size class cannot even be *evaluated* on another—at odds with the amortization argument that justifies learning in the first place.

One question we do not reopen. Intervals admit no natural total order, so any comparison of interval makespans embeds a modelling choice; we adopt the lexicographic ranking that a systematic study of the alternatives found most robust for this problem (Díaz, González-Rodríguez, Palacios, Díaz, & Vela, 2020), and keep it fixed throughout so that every method in this paper is judged by the same criterion.

This paper addresses both obstacles and asks three questions. *Q1*: can a constructive policy trained on a handful of interval instances compete with strong hand-crafted constructive baselines? *Q2*: can a single network, trained on one size class, remain useful across the whole benchmark without retraining? *Q3*: which standard ingredients of the JSP-DRL toolbox—attention encoders, multi-size training, automatic hyperparameter configuration—actually pay off at practical training budgets?

### *Contributions.*

1. **A size-invariant, uncertainty-aware policy for the IJSP** (Section 4). All features are dimensionless: temporal quantities are scaled by a per-instance lower bound and interval uncertainty enters as relative widths. A Deep Sets (Zaheer et al., 2017) encoder over the eligible-operation set, combined with a pooled global context, yields policy *and* value networks of  $\approx 1.2 \times 10^5$  parameters, independent of the numbers of jobs and machines.
2. **An empirical characterization** (Section 6) answering Q1–Q3: budget scaling on the training class, zero-shot dominance over the best dispatching rule on all seven Taillard size classes from a single network, and transfer to a second benchmark of classical instances.
3. **A comparison against both families of constructive heuristic** (Section 6.4), on the 70 benchmark instances and with per-instance results reported in full (Appendix A). The hand-crafted side is the traditional dispatching rules—SPT, LPT, MOR, MWKR and EST, plus two Giffler–Thompson variants—which a single greedy pass improves on instance by instance. The learned-symbolic side is a rule evolved by genetic programming for this same benchmark (Díaz Rodríguez, 2026b), a comparison Xu et al. (2025) single out as missing from the literature: the two paradigms are rarely measured against each other under a common protocol. Here the answer is budget-dependent rather than a verdict—pass for pass the evolved rule wins, and at 1024 samples each the policy does—which is itself the informative outcome.
4. **An analysis of what earns its cost** (Section 7): a permutation audit of which inputs the network consults, six controlled ablations (inference budget, attention versus pooling, specialist versus multi-size training, and three that delimit what interval awareness contributes) of which four are negative results, a Monte Carlo comparison of how faithfully the schedules execute, and a two-campaign automatic-configuration study whose racing optima repeatedly failed to transfer to the operating budget.

Section 2 reviews related work and Section 3 states the problem. Section 4 presents the policy, Section 5 the experimental methodology and Section 6 the results; Section 7 analyses them and states the limitations, and Section 8 concludes.

## 2 Related Work

### 2.1 Job shop scheduling under interval uncertainty

Uncertain processing times have been modelled with probability distributions, fuzzy numbers and intervals. The fuzzy JSP line is mature (González-Rodríguez et al., 2010; Palacios, González-Rodríguez, Vela, & Puente, 2015); the interval model can be read as its minimal-commitment limit, retaining only support bounds. Lei (2011) opened the interval line with population-based neighbourhood search; genetic algorithms with interval rankings (Díaz et al., 2020), robustness-oriented tardiness models (Díaz, Palacios, Díaz, Vela, & González-Rodríguez, 2022), and analytical treatments of special cases (Sotskov, Matsveichuk, & Hatsura, 2020) followed, and interval durations now

appear in richer shop variants as well (Zheng, Xiao, Zhang, & Lv, 2024). On the Taillard-derived benchmark used here, the strongest published results belong to elitist artificial bee colony algorithms (Díaz et al., 2023a, 2023b); these works also fixed the evaluation conventions this paper adopts (interval arithmetic, lexicographic ranking during construction, expected makespan for reporting).

## 2.2 Priority dispatching rules

The oldest constructive approach to the job shop builds a schedule in a single left-to-right pass, repeatedly choosing which of the eligible operations to place next. A *priority dispatching rule* makes that choice by scoring the candidates with a fixed formula over their attributes. The rules this paper uses as baselines and as verification anchors are the classical single-attribute ones, and we keep their usual acronyms. Two look at the operation itself: *shortest* and *longest processing time* (SPT, LPT) prefer the smallest and the largest duration. One looks at the schedule built so far: *earliest starting time* (EST) prefers the operation that could begin soonest given the current occupation of its machine and the progress of its job. Two look at the job the operation belongs to: *most work remaining* and *most operations remaining* (MWKR, MOR) prefer the job with the most pending work, counted in time units and in operations respectively. Under interval durations these attributes are intervals themselves, and candidates are compared with the interval ranking introduced in Section 3. Surveys catalogue well over a hundred variants of such rules (Haupt, 1989; Panwalkar & Iskander, 1977) and recent ones revisit those still in use (Đurašević & Jakobović, 2018).

A refinement worth separating from the rules themselves is the active-schedule generator of Giffler and Thompson (1960), which at each step restricts the candidates to the conflict set of the machine that would finish earliest, and applies a priority rule only to break ties within that set. The restriction guarantees *active* schedules—no operation can be started earlier without delaying another—and lifts the plain rules substantially; we write G&T-SPT and G&T-MWKR for the two combinations used here.

Their appeal is cost: one pass, no search, a solution in milliseconds. Their limit is that a fixed formula cannot suit every shop and objective (Haupt, 1989), which is the opening the learned rules of the next subsection address—and the reason a constructive policy is compared against these rules and not against per-instance metaheuristics, which belong to a different computational class.

## 2.3 Learning constructive heuristics for scheduling

Neural combinatorial optimization began with pointer networks (Vinyals, Fortunato, & Jaitly, 2015) and their RL training (Bello, Pham, Le, Norouzi, & Bengio, 2016); attention encoders became standard for routing (Kool, van Hoof, & Welling, 2019; Nazari, Oroojlooy, Snyder, & Takáč, 2018). For the deterministic JSP, Zhang et al. (2020) learn dispatching policies over disjunctive-graph GNN embeddings, Park et al. (2021) refine such representations and report cross-size generalization, and Tassel et al. (2021) contribute an efficient environment formulation. Sampling several rollouts and

keeping the best is a standard inference-time lever (Kool et al., 2019). Permutation-invariant set encoders were formalized as Deep Sets (Zaheer et al., 2017), of which attention is the natural strict generalization; our architecture starts from the simpler construction and evaluates attention as a controlled ablation.

Learning constructive heuristics is older than its neural incarnation: genetic programming (GP) hyper-heuristics evolve interpretable priority expressions over hand-crafted attributes (Branke, Nguyen, Pickardt, & Zhang, 2016; Nguyen, Mei, & Zhang, 2017). The survey by Xu et al. (2025) compares the GP and RL families for job shop scheduling and notes the scarcity of comparisons under a common protocol. A companion study (Díaz Rodríguez, 2026b) develops the GP side for the interval benchmark used here, which lets Section 6.4 put the two artefacts on the same instances at matched *inference* budgets. A controlled comparison of the two paradigms with *training* budgets matched as well is beyond the scope of either paper and is the subject of dedicated ongoing work.

### 3 Problem Statement

**Definition 1** (IJSP) An IJSP instance comprises  $n$  jobs  $J_1, \dots, J_n$  and  $m$  machines. Job  $J_j$  is a fixed sequence of  $m$  operations  $(o_{j1}, \dots, o_{jm})$ , where  $o_{jk}$  requires a specific machine  $\nu_{jk}$  and an uncertain processing time known to lie in the interval  $\mathbf{p}_{jk} = [p_{jk}^L, p_{jk}^U]$ ,  $0 \leq p_{jk}^L \leq p_{jk}^U$ . Each machine processes at most one operation at a time and preemption is not allowed.

A solution is a processing order of all  $nm$  operations compatible with job precedences; we encode it as a *permutation with repetition* of job indices, where the  $k$ -th occurrence of  $j$  denotes  $o_{jk}$ . Given an order, its *semiactive schedule* starts every operation as early as its job and machine predecessors permit. With component-wise interval addition  $[a, b] + [c, d] = [a + c, b + d]$  and join  $\max([a, b], [c, d]) = [\max(a, c), \max(b, d)]$ , starts and completions are intervals:

$$\mathbf{s}_o = \max(\mathbf{c}_{JP(o)}, \mathbf{c}_{MP(o)}), \quad \mathbf{c}_o = \mathbf{s}_o + \mathbf{p}_o, \quad (1)$$

where  $JP(o)$  and  $MP(o)$  are the job and machine predecessors of  $o$ . The interval makespan collects the job completions component-wise,

$$\mathbf{C}_{\max} = \left[ \max_j c_{o_{jm}}^L, \max_j c_{o_{jm}}^U \right], \quad (2)$$

and, because Eq. (1) is monotone in the durations, the executed makespan of *any* realization of the processing times falls inside  $\mathbf{C}_{\max}$ . Intervals carry no canonical total order, so schedule comparison requires a ranking choice; during construction and search we use the lexicographic order

$$\mathbf{a} \prec_{lex} \mathbf{b} \iff a^U < b^U \vee (a^U = b^U \wedge a^L < b^L), \quad (3)$$

which prioritizes the worst case, in line with min-max robust optimization and with the IJSP literature (Díaz et al., 2020, 2023a). For reporting we follow the field

convention (Díaz et al., 2020, 2023a):

$$\text{RE}(\%) = \frac{E[\mathbf{C}_{\max}] - \text{LB}}{\text{LB}} \times 100, \quad E[\mathbf{C}_{\max}] = \frac{1}{2}(C_{\max}^L + C_{\max}^U), \quad (4)$$

with LB the published lower bound of the crisp Taillard counterpart (Taillard, 1993)—an indicative, hardware-independent yardstick rather than a strict optimality gap, since the crisp JSP and the IJSP are different problems.

*Remark 1* For instances whose intervals are symmetric perturbations of a crisp instance, the midpoint scenario coincides with the original crisp durations; comparisons under Eq. (4) are therefore scenario-consistent (midpoint solution quality against the midpoint-scenario bound).

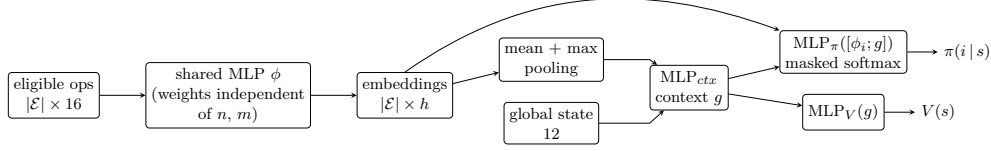
## 4 Method

Reinforcement learning is learning to decide from experience. An *agent* interacts with an environment in steps: it observes the current *state*  $s$ , chooses an *action*  $a$  among those available, and receives a numeric *reward* signalling how good the consequences were. The agent’s behaviour is its *policy*  $\pi(a | s)$ : a function that, given the state  $s$ , assigns a probability to each available action  $a$ . The policy is the object of learning: training increases the probability of actions followed by a high cumulative reward (the *return*).

Policy-gradient methods such as PPO (Schulman, Wolski, Dhariwal, Radford, & Klimov, 2017) judge an action not by its return alone but by comparing that return with what was *expected* from the state where the action was taken; only the surplus—the *advantage*—moves the policy. The expectation is itself learned: a *value function*  $V(s)$  estimates the return attainable from state  $s$ . In practice both functions share one network with two outputs, a *policy head* that scores the available actions and a *value head* that estimates  $V(s)$ , trained jointly. This is why the network in this paper has two heads.

Instantiated on scheduling, the agent is a dispatcher. The state is the partial schedule; the available actions are the *eligible* operations—the next unscheduled operation of each job, so choosing an action  $a$  means choosing which eligible operation  $i$  to schedule next, and we write the policy over candidates as  $\pi(i | s)$ ; an *episode* is the construction of one complete schedule, and the reward rests on the makespan achieved. The trained policy then plays the same role as the priority dispatching rules of Section 2.2: where a hand-crafted rule orders the eligible operations by a fixed criterion, the policy orders them by preferences it has learned from experience. Everything around that choice (the schedule builder of Eq. (1), the interval arithmetic, the evaluation) is the same machinery every baseline in this paper uses.

Figure 1 shows the network, read left to right. Each eligible operation enters as a row of 16 numbers that describe it: how long it may take, how much work its job still carries, how soon it could start (Section 4.2). A small multilayer perceptron (MLP)—the *same* one for every candidate—turns each row into an *embedding*: a vector of  $h$



**Fig. 1** The size-invariant network. Every eligible operation is embedded by the same MLP  $\phi$ ; mean and max pooling collapse the variable-size candidate set into fixed-size summaries that, joined with the global features, form the context  $g$ . The policy head scores each candidate embedding against  $g$ ; the value head reads  $g$  alone. No dimension depends on the numbers of jobs or machines.

learned quantities that re-describes the candidate in terms the network finds useful. The number of candidates varies from step to step, so the embeddings are *pooled*: their mean and their componentwise maximum are two fixed-size summaries of the whole set (the typical candidate and the extreme one). Joined with 12 numbers describing the shop as a whole (the *global state*), these summaries are condensed by another MLP into a single *context* vector  $g$ —the network’s picture of the current situation. The two heads of the previous paragraph read from here. The policy head looks at each candidate’s embedding next to  $g$  and scores it; a *softmax* turns the scores into the probabilities  $\pi(i | s)$  (*masked* means that unused padding slots are excluded). The value head reads  $g$  alone and outputs  $V(s)$ . In total, four small two-layer MLPs—the candidate encoder  $\phi$ , the context combiner, and the two heads—composed into a single network and trained jointly end to end.

The rest of the section climbs the same ladder in full detail, from the decision problem to the network. Section 4.1 casts schedule construction as the decision process the agent faces (states, actions, rewards); Section 4.2 defines what the network sees—the 16+12 input quantities, which is where the interval uncertainty and the size invariance enter; Section 4.3 specifies the three network stages just toured; and Section 4.4 presents the self-attention variant evaluated against this base design. How the network is trained, and how it is used once trained, belongs with the experimental setting and is described in Section 5.4.

## 4.1 Dispatching as a Markov decision process

Cast in the terms above, schedule construction is a finite-horizon Markov decision process of exactly  $nm$  steps. The *state* after  $t$  steps comprises the partial schedule (job pointers, job and machine completion intervals) and the set  $\mathcal{E}_t$  of eligible operations—the next unscheduled operation of each unfinished job, so  $1 \leq |\mathcal{E}_t| \leq n$ . The *action* selects one element of  $\mathcal{E}_t$ ; the transition applies Eq. (1). Episodes always terminate after  $nm$  steps.

The *reward* is a weighted sum of six shaping components, each collapsing intervals by their worst case (upper bound); component scales adapt automatically to each instance through its lower bound, and the weights were chosen on the development set of Section 5.1:

**Table 1** Per-operation features (all dimensionless).  $d = \mathbf{p}_{jk}$ ,  $\mathbf{est}$  = earliest start interval,  $\mathbf{rem}$  = remaining work interval of the job after  $o$ ;  $\text{mid}(\cdot)$  is the interval midpoint;  $L(\mu)$  the remaining worst-case load of machine  $\mu$ ;  $\bar{L}$  its average over machines.

| #     | Feature               | Definition                                                               |
|-------|-----------------------|--------------------------------------------------------------------------|
| 1–2   | duration bounds       | $d^L/\text{LB}, d^U/\text{LB}$                                           |
| 3     | duration uncertainty  | $(d^U - d^L)/\text{mid}(d)$                                              |
| 4–5   | earliest start bounds | $\text{est}^L/\text{LB}, \text{est}^U/\text{LB}$                         |
| 6     | start uncertainty     | $(\text{est}^U - \text{est}^L)/\text{mid}(\mathbf{est})$                 |
| 7–8   | remaining job work    | $\text{rem}^L/\text{LB}, \text{rem}^U/\text{LB}$                         |
| 9–10  | job position          | $(m - k - 1)/m, k/m$                                                     |
| 11    | machine load          | $L(\mu)/\text{LB}$                                                       |
| 12    | potential idle gap    | $\max(0, c_J^U - c_\mu^U)/\text{LB}$                                     |
| 13    | machine congestion    | $L(\mu)/\bar{L}$                                                         |
| 14    | slack                 | $(\text{est}^U - \min_{o' \in \mathcal{E}} \text{est}_{o'}^U)/\text{LB}$ |
| 15–16 | completions           | $c_\mu^U/\text{LB}, c_J^U/\text{LB}$                                     |

- *terminal makespan* (weight 1.0): the negative scaled makespan at episode end, plus a bonus when the schedule lands within 5% of the instance bound—the quantity actually being minimized;
- *local improvement* (weight 0.3): the step change of the projected makespan, penalizing deteriorations twice as much as it rewards improvements—a dense signal so early decisions are not credited only  $nm$  steps later;
- *machine idleness* (0.15) and *load balance* (0.15): discourage gaps on machines and lopsided load, the classic proxies for a compact schedule;
- *operation criticality* (0.05) and *job progress* (0.05): small nudges towards operations on the emerging critical path and towards finishing what is started.

## 4.2 Dimensionless, uncertainty-aware features

Raw times do not travel between instances: a duration of 80 is long when the whole schedule spans 1,200 time units and short when it spans 3,000, so a network fed raw values would be tied to the scale it was trained on. The representation therefore expresses every temporal quantity as a fraction of the instance’s lower bound LB (worst case if the bound is itself an interval), and every uncertainty as a ratio of interval width to magnitude. What the policy sees is thereby comparable across instances—and across sizes, which is half of the size invariance. Each eligible operation  $o = o_{jk}$  on machine  $\mu = \nu_{jk}$  is described by the 16 dimensionless quantities of Table 1; the global context is the 12-dimensional aggregate of Table 2. Two design rules follow from ablation evidence (Section 7.4): (i) features must add information that is *not* a linear combination of others (interval widths enter as relative, hence non-linear, magnitudes); and (ii) normalization must be co-designed with initialization and learning rates rather than retrofitted.



**Table 2** Global context features (all dimensionless aggregates; no dimension depends on  $n$  or  $m$ ).

| Feature             | Definition                                                          |
|---------------------|---------------------------------------------------------------------|
| progress            | scheduled operations / $nm$                                         |
| partial makespan    | $\max_{\mu} c_{\mu}^U / \text{LB}$                                  |
| machine completions | mean and std of $c_{\mu}^U / \text{LB}$                             |
| remaining loads     | mean, max and std of $L(\mu) / \text{LB}$                           |
| remaining job work  | mean and max of job remaining / $\text{LB}$                         |
| eligibility         | $ \mathcal{E} /n$                                                   |
| pending uncertainty | $\sum (\text{widths}) / \sum (\text{midpoints})$ of unscheduled ops |
| total load          | $\sum_{\mu} L(\mu) / (\text{LB} \cdot m)$                           |

### 4.3 Size-invariant architecture

The remaining obstacle is that the number of candidates changes at every step and from instance to instance, while a neural network wants inputs of a fixed size. The architecture dissolves this with two ideas: the *same* small network reads every candidate, so any number of them can be processed; and the resulting embeddings are *pooled* into fixed-size summaries, so the scoring stage can see the whole decision at once. Formally, let  $x_i \in \mathbb{R}^{16}$  be the features of eligible operation  $i$  and  $g_0 \in \mathbb{R}^{12}$  the global features; the construction has three stages (Figure 1).

*Embed.* The shared two-layer MLP  $\phi$  (width  $h=128$ , LayerNorm, ReLU, orthogonal initialization) maps each candidate to an embedding  $\phi_i = \phi(x_i)$ . Because the same weights process every candidate, the layer accepts any number of them—this is where size invariance over jobs comes from.

*Pool into a context.* The variable-size set  $\{\phi_1, \dots, \phi_{|\mathcal{E}|}\}$  must be summarized into a fixed-size vector before anything global can be computed. Mean and max pooling provide two complementary permutation-invariant summaries (Zaheer et al., 2017)—the average candidate and the extreme one—which, concatenated with the global features  $g_0$ , are mixed into the context:

$$g = \text{MLP}_{ctx}([\frac{1}{|\mathcal{E}|} \sum_i \phi_i; \max_i \phi_i; g_0]) \in \mathbb{R}^h. \quad (5)$$

This embed-then-pool pattern is the *Deep Sets* construction of Zaheer et al. (2017), whose central result is that *any* function invariant to the order of a set can be written in this form—so the simplicity costs no expressiveness in principle. The result tables refer to the base model by this name, in contrast to its attention variant (Section 4.4).

*Score.* The *policy head* rates each pair (candidate embedding, context)—“how good is operation  $i$  given the decision as a whole”—and a softmax over the true candidates (padding masked out) turns the scores into probabilities. The *value head* estimates the return attainable from the current state—the  $V(s)$  of the advantage computation introduced above—and reads the context alone, since expectations concern the

situation, not any one candidate:

$$\pi(i | s) = \text{softmax}_i \text{MLP}_\pi([\phi_i; g]), \quad V(s) = \text{MLP}_V(g). \quad (6)$$

No parameter dimension depends on  $n$  or  $m$ : the same  $\approx 1.2 \times 10^5$ -parameter network dispatches any instance. The last policy layer is initialized with gain  $10^{-2}$ , so at the start every candidate receives a nearly equal score and the untrained policy is close to uniform.

#### 4.4 The attention variant

Pooling summarizes the candidate set by two statistics, its average and its extreme, and every candidate is then scored against that same summary. What this cannot express is a comparison between two *particular* candidates: whether operation  $i$  deserves the machine now may depend on which other eligible operation would claim it next, and a mean says nothing about which candidate that is. *Self-attention* is the standard remedy, and the ingredient the neural combinatorial optimization literature reaches for first (Section 2.3). In an attention layer every candidate emits a *query*, a *key* and a *value*; the similarity between one candidate’s query and another’s key decides how much of that other’s value is added into the first’s embedding. Each candidate is thereby rewritten as a content-weighted reading of all the others, with the weights computed from the embeddings themselves rather than fixed in advance.

We build this as a variant to be tested, not as part of the proposed model. A stack of  $B$  pre-layer-normalization transformer blocks transforms  $\{\phi_i\}$  after the shared encoder and before pooling, that is, between the embedding and pooling stages of Figure 1. Each block normalizes the embeddings, applies multi-head attention over them (4 heads) and adds the result back, then applies a two-layer feed-forward network the same way; both additions are *residual*, that is, the block adds to its input instead of replacing it, so an untrained block behaves almost like the identity and the stack starts close to the base model. No positional encoding is used: the candidates form a set with no meaningful order, and numbering them would destroy the permutation invariance Section 4.3 was built to guarantee. Padding slots are masked out of the attention weights, so no candidate ever attends to an operation that is not there.

Attention is a flag on the same code path, and  $B=0$  reproduces the base network exactly—which is what makes the comparison of Section 7.3 a controlled one, differing in this component and nothing else. Its price is not small. A single block carries  $\approx 1.3 \times 10^5$  parameters—the query, key and value projections, the output projection, two layer normalizations and the feed-forward network—so the  $B=2$  variant we evaluate grows the network from  $1.2 \times 10^5$  to  $3.9 \times 10^5$  parameters, a factor of 3.2, and makes an episode about 30% more expensive. Section 7.3 reports what that buys.

During batched training, variable-size candidate sets are padded to the batch maximum with a boolean mask applied to attention, pooling and logits; unit tests verify that masked-padded forward passes are numerically identical to unpadded ones.

**Table 3** Machine and software used for every experiment in this paper.

| Component        | Specification                                   |
|------------------|-------------------------------------------------|
| CPU              | AMD Ryzen 5 3400G, 4 cores / 8 threads, 3.7 GHz |
| Memory           | 32 GB                                           |
| GPU              | NVIDIA GeForce RTX 5060 Ti, 16 GB (tuning only) |
| Operating system | Windows 10 Pro 22H2 (64-bit)                    |
| Language         | Python 3.12.6                                   |
| Learning         | PyTorch 2.9.1, CUDA 13.0 build                  |
| Numerics         | NumPy 2.3.5, SciPy 1.17.0                       |
| Figures          | Matplotlib 3.10.8                               |
| Tuning           | irace 4.4.3 on R 4.6.1 (Section 5.3)            |

## 5 Experimental Methodology

### 5.1 Instances, data split and metric

Experiments use the 70 interval Taillard instances TA1–TA70 that anchor the recent IJSP literature (Díaz et al., 2023a, 2023b): seven size classes from  $15 \times 15$  to  $50 \times 20$ , each crisp duration replaced by an integer interval symmetric around it, with half-width drawn uniformly up to 15% of the original value (Díaz et al., 2020). The benchmark is openly available (Díaz Rodríguez, 2026a). Of the 70 instances, four carry gradients: the final model trains on **TA11–TA14** ( $20 \times 15$ ) only. Six more of the same class, **TA15–TA20**, form the *development set*—never used for gradients, but every design and configuration decision was judged on their performance, so they are reported separately and excluded from claims about unseen data. The remaining 60 instances were touched by neither training nor development. Three independent training runs, differing only in their random seed, are reported as mean and best. Performance is measured by RE, Eq. (4), using the published crisp lower bounds.

### 5.2 Computational environment

Table 3 lists the machine and the software stack. Two remarks matter for reading the costs reported later. First, every number in this paper—training, evaluation, the dispatching-rule baselines and the ablations—was produced on the CPU. The network is small enough (Section 4.3) that a forward pass is negligible next to the environment step that follows it, so the GPU buys nothing for a single rollout; it was used only for the vectorized batched rollouts of the operating-fidelity tuning campaign (Section 5.3), where many episodes advance in lockstep and the batched forward pass does pay. The method therefore carries no dependence on specialized hardware, which is part of what makes the deployment argument of Section 6.4 credible.

Second, the wall-clock figures quoted there are measurements on this machine, whose four physical cores were at times shared by concurrent experimental arms. They are reported as orders of magnitude—hours to train, milliseconds to dispatch—and not as a timing benchmark; nothing in the paper’s conclusions turns on them.

**Table 4** Hyperparameters, identical across every experiment reported in this paper. Most are standard PPO values; the discount factor and the batched update scheme depart from common practice for the reasons given in Section 5.4. None was tuned for this problem, though not for want of trying: two irace campaigns searched the space of Table 5 for better ones and neither transferred to the operating budget.

| Hyperparameter              | Value             | Hyperparameter           | Value      |
|-----------------------------|-------------------|--------------------------|------------|
| <i>Optimizer</i>            |                   | <i>PPO objective</i>     |            |
| learning rate (Adam)        | $3 \cdot 10^{-4}$ | clipping range           | 0.2        |
| Adam $\epsilon$             | $10^{-5}$         | GAE $\lambda$            | 0.95       |
| gradient clipping           | 0.5               | discount factor $\gamma$ | 1.0        |
| <i>Network and schedule</i> |                   | <i>PPO updates</i>       |            |
| hidden width $h$            | 128               | epochs per update        | 4          |
| weight initialization       | orthogonal        | minibatch size           | 256        |
| policy update               | every 4 ep.       | entropy coefficient      | 0.01       |
| greedy evaluation           | every 10 ep.      | advantage normalization  | per update |

**Table 5** Search space of the two irace campaigns. Real parameters were sampled on a logarithmic scale, the rest from the listed values. Every default of Table 4 lies inside the space, so the race could have returned it. The second campaign fixed minibatch size and update period at their defaults, on which the elites of the first had agreed unanimously.

| Parameter           | Default           | Campaign 1 (330 exp.) | Campaign 2 (300 exp.) |
|---------------------|-------------------|-----------------------|-----------------------|
| learning rate       | $3 \cdot 10^{-4}$ | $[10^{-4}, 10^{-3}]$  | $[10^{-4}, 10^{-3}]$  |
| entropy coefficient | 0.01              | $[0.003, 0.03]$       | $[0.003, 0.03]$       |
| clipping range      | 0.2               | 0.1, 0.2, 0.3         | 0.1, 0.2, 0.3         |
| epochs per update   | 4                 | 2, 4, 8               | 4, 8                  |
| GAE $\lambda$       | 0.95              | 0.90, 0.95, 0.98      | 0.90, 0.95, 0.98      |
| hidden width        | 128               | 64, 128, 256          | 128, 256              |
| minibatch size      | 256               | 128, 256, 512         | fixed                 |
| update period       | 4 ep.             | 2, 4, 8 ep.           | fixed                 |

### 5.3 Hyperparameter configuration

All reported results use the hyperparameters of Table 4 *without problem-specific tuning*. To assess whether this leaves performance on the table, we ran an automatic configuration study with irace (López-Ibáñez, Dubois-Lacoste, Pérez Cáceres, Birattari, & Stützle, 2016). irace *rac*es candidate configurations: they are evaluated side by side on a growing set of instances (here, training seeds), statistical losers are dropped early, and the budget concentrates on the survivors, called the *elite* configurations. The race covered the 8-parameter space of Table 5 over a budget of 330 tuning experiments (309 raced) at reduced fidelity (1 training instance, 300 episodes,  $\sim 8$  minutes per evaluation), with training seeds as racing instances. The five elite configurations agreed on a coherent faster-learning regime (learning rate  $7\text{--}9 \cdot 10^{-4}$ , clip 0.3, minibatch 128, updates every 2 episodes,  $\lambda=0.90$ ) and were insensitive to network width.

However, at the full 1000-episode operating budget the best elite scored 14.5% mean RE versus 13.4% for the defaults (+1.66% mean makespan; five of six development instances within noise, one significantly worse) at  $\sim 60\%$  higher training cost: the low-fidelity optimum did not transfer.

To rule out that this non-transfer was an artifact of the reduced tuning fidelity, a second, *operating-fidelity* campaign raced 295 experiments of a 300 budget, each training at the full 1000 episodes on two instances (enabled by a  $\sim 3\times$  GPU-vectorized rollout implementation), over the reduced 6-parameter space of Table 5, seeded with the defaults and the low-fidelity elites. This campaign converged to a coherent region *distinct* from the defaults (clip 0.1, 8 epochs,  $\lambda=0.90$ , width 256, learning rate  $5-9\cdot 10^{-4}$ ) and eliminated the seeded default during racing. Yet when its winning configuration was retrained on the full four-instance set over three seeds and evaluated against the default checkpoints under an identical protocol, it again failed to improve: 14.73% versus 13.52% mean RE over the six development instances (+1.21 points; better on two instances, worse on four). Two conclusions follow: automatic configuration does not beat the defaults of Table 4 for this problem *even when performed at the operating budget*, so the reported results do not depend on fine tuning; and the recurring gap between racing-optimal and confirmation-optimal configurations is itself a cautionary observation for automatic DRL configuration, where single-run training cost is too noisy to separate near-equivalent configurations.

## 5.4 Training and inference

We train with PPO (Schulman et al., 2017); Table 4 lists all hyperparameters. Two choices depart from common JSP-DRL practice and are enabled by co-design with the normalized representation: a discount factor  $\gamma = 1$  (undiscounted finite horizon—future rewards count in full; with the customary  $\gamma = 0.99$  the terminal makespan signal reaches early decisions attenuated by  $0.99^{nm} \approx 0.05$  for  $20\times 15$  instances) and batched minibatch updates with advantage normalization. Training on a set of instances proceeds in blocks (one instance at a time) with full weight transfer. A *rollout* is one complete construction episode played by the current policy: *sampled* if each operation is drawn from  $\pi$ , as during training, or *greedy* if every step takes the most probable one. Sampled training episodes are interleaved with periodic greedy evaluations, and the best solution and best saved weights (the *checkpoint*) are tracked across both.

### *Best-of- $N$ inference.*

At evaluation time we exploit the stochastic policy: one greedy rollout plus  $N - 1$  sampled ones, keeping the best solution under the ranking of Eq. (3) ( $N=64$ ). This choice is motivated empirically in Section 7.2: during training, the best solutions frequently originate from sampled episodes, i.e. the policy is more valuable as a *distribution* than as its argmax—consistent with sampling-based decoding in neural combinatorial optimization (Kool et al., 2019).

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**Algorithm 1** Best-of- $N$  inference

---

```
1:  $(\sigma^*, \mathbf{C}_{\max}^*) \leftarrow$  greedy rollout of  $\pi$ 
2: for  $i = 1$  to  $N - 1$  do
3:    $(\sigma, \mathbf{C}_{\max}) \leftarrow$  sampled rollout of  $\pi$ 
4:   if  $\mathbf{C}_{\max} \prec_{lex} \mathbf{C}_{\max}^*$  then  $(\sigma^*, \mathbf{C}_{\max}^*) \leftarrow (\sigma, \mathbf{C}_{\max})$ 
5:   end if
6: end for
7: return  $\sigma^*$ 
```

---

## 5.5 Baselines

Constructive baselines are the dispatching rules of Section 2.2—SPT, LPT, MOR and MWKR, of which MOR is uniformly the strongest, plus EST—and the two Giffler–Thompson combinations. G&T-MWKR is the strongest deterministic constructive baseline on this benchmark: 29.5% mean RE across the 70 instances, versus 42.3% for EST—the strongest plain rule overall, though weaker than MOR on the  $20 \times 15$  training class—and 45.5% for MOR (the SPT tie-break variant performs poorly, 70.6%). Published IJSP metaheuristics (fEABC (Díaz et al., 2023b) and ESABC (Díaz et al., 2023a); 30 runs and minutes of per-instance search on dedicated hardware) are included as reference yardsticks from a different computational class rather than as direct competitors.

## 6 Results

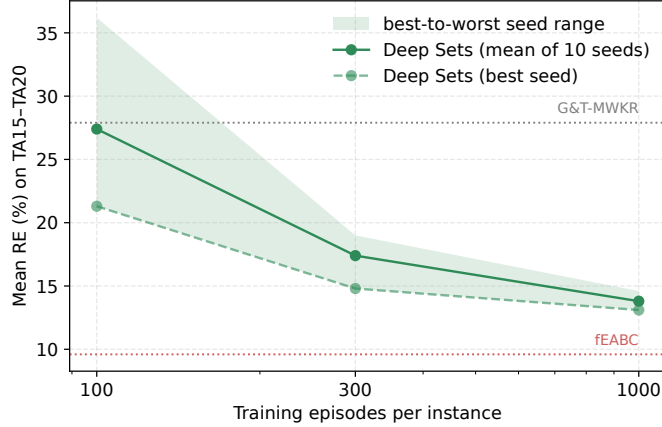
### 6.1 In-size performance and budget scaling

Table 6 reports per-instance RE on the development set for increasing training budgets over ten independent training runs, and Figure 2 the aggregate picture. Three observations: (i) the policy scales monotonically with budget (27.4%  $\rightarrow$  17.4%  $\rightarrow$  13.8% mean RE at 100/300/1000 episodes per instance) with clearly diminishing returns at the last step, locating the pure-budget ceiling near 11–12%; (ii) the spread between training runs collapses as the budget grows, the standard deviation across the ten seed means falling from 4.69 points at 100 episodes to 1.30 at 300 and 0.47 at 1000—a factor of ten—so what the extra budget buys is not only a better mean but a reproducible one; (iii) the gap to the best dispatching rule is a factor  $\approx 3.4$ .

The third observation is the one that needed the extra runs. Reported over three seeds, a collapse in variance is difficult to distinguish from three runs that happened to agree. Over ten, the worst seed at 1000 episodes reaches 14.62% and the best 13.12%, a range narrower than the gap between any two budgets, whereas at 100 episodes the same ten span 21.3 to 36.2%. Ablations elsewhere in this paper are paired on the first three of these seeds, whose mean is 13.44%; that figure is the baseline those comparisons subtract, not a second estimate of the policy’s performance.

**Table 6** RE (%) per development instance, mean [best] over ten training runs differing only in their random seed, Deep Sets model, best-of-64 inference. The rightmost column is the standard deviation across the ten per-seed means over the six instances.

| Budget   | TA15               | TA16               | TA17               | TA18               | TA19               | TA20               | Mean        | SD          |
|----------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-------------|-------------|
| 100 eps  | 30.2 [21.9]        | 26.3 [19.7]        | 26.8 [18.1]        | 26.7 [18.4]        | 28.7 [21.5]        | 25.8 [18.5]        | 27.4        | 4.69        |
| 300 eps  | 19.0 [16.3]        | 16.3 [14.3]        | 16.6 [13.3]        | 18.5 [16.2]        | 17.2 [13.3]        | 16.6 [14.9]        | 17.4        | 1.30        |
| 1000 eps | <b>14.9</b> [12.1] | <b>13.2</b> [11.8] | <b>12.9</b> [10.4] | <b>16.3</b> [14.6] | <b>12.8</b> [11.1] | <b>12.8</b> [10.5] | <b>13.8</b> | <b>0.47</b> |
| MOR      | 51.3               | 35.8               | 50.1               | 54.2               | 43.2               | 43.7               | 46.4        | —           |



**Fig. 2** Mean RE on the development set versus training episodes per instance, over ten training runs; the band spans the best and worst of them, and its collapse is observation (ii). Dotted lines: the strongest constructive baseline, G&T-MWKR, and fEABC as a yardstick from a different computational class, both on these same six instances. The attention variant is not drawn here—it has three seeds against these ten, so the paired comparison belongs in Table 10.

## 6.2 Zero-shot cross-size generalization

The size-invariant design allows direct evaluation of the  $20 \times 15$ -trained network on any class. Evaluation is *zero-shot*: the trained weights are applied to sizes they never saw, with no additional training or fine-tuning of any kind. Table 7 reports zero-shot RE over all seventy instances, ten per class with none omitted: every seed’s network outperforms MOR on *every* one of them, with graceful degradation (class means from 11.0% on  $15 \times 15$  to 22.0% on the hardest transfer,  $30 \times 20$ ) and no failure mode up to 1000-operation episodes. Degradation is not monotone in instance size:  $50 \times 15$ , at 12.2%, transfers better than either 30-machine class, which the dispatching rules also find easier, so the difficulty the network inherits is the instances’ and not a limit of the transfer. Transfer quality improves with in-size training budget (mean over seeds:  $30 \times 20$  from 30.3% with the 300-episode checkpoints to 25.9% with the 1000-episode ones;  $50 \times 15$  from 22.2% to 15.3%), i.e. the *specialist*—our shorthand for the network trained on the single  $20 \times 15$  class—learns structure that is not size-bound.

**Table 7** Zero-shot RE (%) of the  $20 \times 15$ -trained network (best-of-64), mean [best] over the three training seeds, by size class over all 70 instances—ten per class, none omitted. The  $20 \times 15$  row (†) contains the training and development instances and is not a generalization result; the other six classes were never trained on. Per-instance values are in the appendix.

| Class                  | Policy             | MOR  | G&T-MWKR |
|------------------------|--------------------|------|----------|
| $15 \times 15$         | 11.0 [9.6]         | 41.4 | 26.7     |
| $20 \times 15^\dagger$ | 13.6 [13.0]        | 46.7 | 29.0     |
| $20 \times 20$         | 14.8 [14.0]        | 47.3 | 30.9     |
| $30 \times 15$         | 16.8 [16.0]        | 44.1 | 33.6     |
| $30 \times 20$         | 22.0 [20.7]        | 58.8 | 36.8     |
| $50 \times 15$         | 12.2 [11.1]        | 34.1 | 23.9     |
| $50 \times 20$         | 15.5 [13.9]        | 45.9 | 25.5     |
| All                    | <b>15.1</b> [14.0] | 45.5 | 29.5     |

### 6.3 Cross-family generalization: the classical instances

The IJSP literature also reports on interval versions of twelve classical instances (FT10 and FT20, seven of the La family, and ABZ7–9), generated from their crisp counterparts with the same symmetric scheme as the Taillard set (Díaz et al., 2020). None of these families was seen during training or development, so they double as a cross-family generalization test. Following the literature, RE is measured here against the best-known expected-makespan values (Díaz et al., 2023b)—a tighter reference than Taillard’s crisp bounds, so Table 8 is not comparable with the Taillard tables.

Table 8 positions the policy (best-of-64, mean and best of the three seeds) against the shared-evaluator rules and the published 30-run averages of the per-instance meta-heuristics. The policy improves on EST and G&T-MWKR on all twelve instances and lands at 12.4% mean RE—between the hand-crafted constructive baselines (30.1% for G&T-MWKR) and the per-instance searchers (GA at 9.8%, ESABC at 5.5%). On La40, ABZ7 and ABZ8 its best seed edges past the GA average itself, with no retraining of any kind.

Against the evolved rules the comparison must be made budget by budget, as it was on the Taillard set, and Figure 3 is arranged that way. Given one pass each, the policy’s 17.2% leads the 19.0% a single evolution should be expected to yield, and trails the featured rule’s 17.9% on 5 of the 12 instances (Wilcoxon signed-rank against the featured rule,  $p = 0.73$ : the two are indistinguishable). Given 64 samples each—the rules randomized by the  $\epsilon$ -greedy dispatching their own study provides—the policy leads on both readings, 12.4% against 14.6% for the mean of 30 and 14.2% for the featured rule, better than the latter on 9 of 12, though on twelve instances even that is only marginal ( $p = 0.077$ ). At 1024 samples per run the gap is both the widest and the only one that separates: 10.3% against 12.2%, better on 10 of the 12 instances ( $p = 0.009$ ). Reading these three rows in order, the policy is level with the rules at one pass and pulls away as both are allowed to sample—which is what



Section 6.4 finds on the Taillard set, arrived at here on a family neither method was tuned for. What separates the network from the formula is not the decision it makes but the distribution it samples from.

The 1024 figures deserve one note about protocol. Elsewhere this paper spends that budget as 342 samples on each of the three checkpoints, keeping the best across them, which is what the deployed system does and what Section 6.4 compares. Here each checkpoint receives its own 1024 and the three are averaged, because a single GP run is a single rule and pooling three networks against one rule would match the sample budget while unmatching the model count. The two protocols land in the same place, 10.13% pooled against 10.27% per run, so nothing turns on the choice; it is made explicit because the reader cannot check it otherwise.

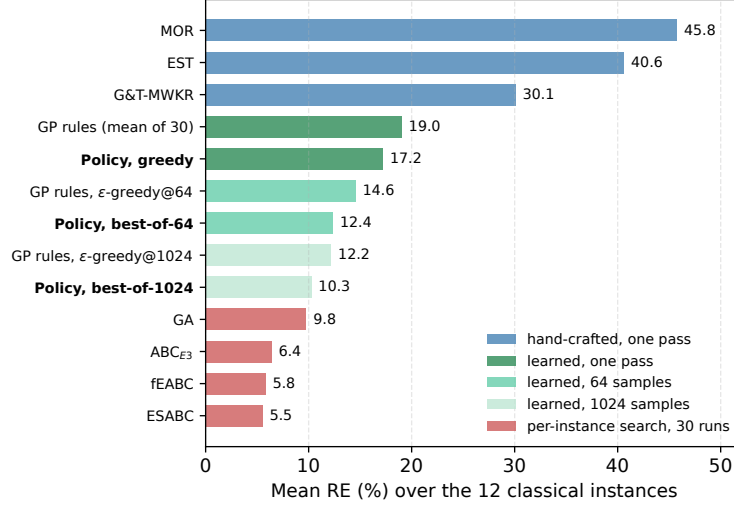
Table 8 reports both learners as a mean over independent runs, which changes the comparison in the policy’s favour and is the fairer statement. Against what a single evolution should be expected to yield—19.0%, 14.6% and 12.2% at the three budgets—the policy’s 17.2%, 12.4% and 10.1% lead at every one. Against the best of the 30 rules, which averages 15.3%, 12.8% and 11.0%, it trails at one pass and leads at the other two. Both readings are in the table because the choice between them is the reader’s: one asks what the method delivers, the other what its luckiest run delivered.

The published rule sits between them, and not by accident. That study selects it as the lowest mean RE *on the 70 Taillard instances*, which is a best-of-30 chosen on data the twelve classical instances play no part in; its 17.9%, 14.2% and 11.3% beat the 30-run mean at all three budgets, so the selection does transfer across families—but only partway, since it lands at the fortieth percentile of the 30 here rather than at the top.

The spread is the more useful observation. Per instance the standard deviation across the 30 evolved rules averages 4.3 points, against 2.2 across the policy’s three seeds. Neither figure should be pushed far, since 30 runs and 3 seeds are not the same evidence, but the direction matters for anyone choosing between the two approaches: evolving one rule and training one network do not expose a practitioner to the same risk from the run they happen to get.

## 6.4 Position among constructive methods

Training the final model takes 66–123 minutes per seed across the three seeds ( $4 \times 1000$  episodes) on the commodity CPU of Table 3. The cost is dominated by environment interaction, not by the small network: a seed dispatches 1.2 million operations, each paying for feature extraction over the unscheduled work, and the GPU-vectorization of rollouts in Section 5.3 bought its  $\sim 3 \times$  precisely there. Constructing a solution takes milliseconds (greedy) to a few seconds (best-of-64 on  $50 \times 20$ ). Against the hand-crafted rules the comparison is one-sided: a single greedy pass improves on MOR, EST and G&T-MWKR on each of the 70 instances (paired Wilcoxon signed-rank test,  $p < 10^{-12}$ ; the matched-pairs rank-biserial correlation, an effect size that reaches 1 only when one side wins every single pair, is exactly 1). Raising the inference budget to best-of-1024 brings the full-benchmark mean RE to 13.0% over the 70 instances (Table 9 and Figure 5)—ahead of every deterministic constructive baseline on every instance,



**Fig. 3** Mean RE over the 12 classical interval instances, grouped by the inference budget each method is given, since comparing one learner against the other across budgets would confound the method with what it is allowed to spend. Both learned entries are means over independent runs—30 evolutions for the rules, three training seeds for the policy—rather than a selected artefact against a mean.

again with rank-biserial correlation 1, and within 2.5–5.1 points of the fEABC 30-run class averages.

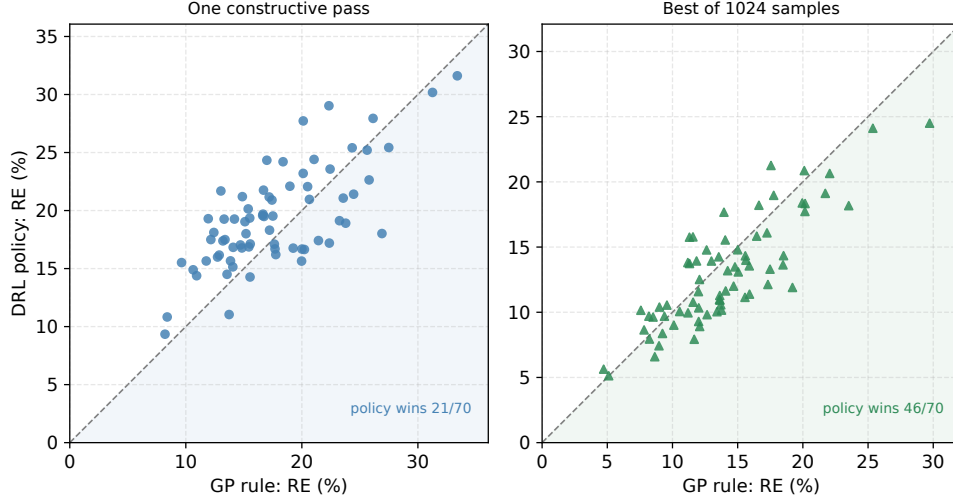
Against the evolved rule of the companion study the comparison is not one-sided, and its answer depends on the inference budget the two are given (Figure 4). Pass for pass, the rule wins: one greedy pass of the policy improves on it in only 21 of the 70 instances, a median deficit of 2.0 points ( $p = 4.6 \times 10^{-4}$ ), and averages 19.4% against the rule’s 17.7%. Which side of that pair is a maximum matters: the 17.7% is the best of the companion’s 30 evolutions, selected on these same 70 instances, whereas the 19.4% is a mean over three seeds. Measured instead against what a single evolution should be expected to yield—18.99%, the 30-run mean that study reports—the deficit at one pass is 0.4 points rather than 1.7. We keep the featured rule as the opponent, since it is the artefact the study publishes and the stronger test, but the reader should know it is a selected one. Given 1024 samples each—the companion randomizes its rule with  $\epsilon$ -greedy dispatching for exactly this purpose—the ordering reverses, though by less than the two headline averages alone would suggest: the policy wins 46 of 70 with a median advantage of 1.3 points ( $p = 1.4 \times 10^{-3}$ ), 13.0% against 14.1%.

The two halves say the same thing from opposite sides. A single decision sequence drawn greedily from  $1.2 \times 10^5$  parameters is worse than a compact evolved formula, so the network’s advantage does not lie in its argmax; what it provides, and the formula does not, is a distribution whose samples improve faster with budget—the reading the best-of- $N$  ablation of Section 7.2 reaches from within. Both learners saw the same four training instances, but under different regimes and costs, so this compares published artefacts rather than a controlled study; the latter, with training as well

**Table 8** RE (%) on the 12 classical interval instances (cross-family zero-shot). Column groups are inference budgets, so that the two learned methods—GP, the evolved rules of the companion study (Díaz Rodríguez, 2026b), and Pol., this policy—are read against each other within a group and never across groups. At 64 samples the rule is randomized by  $\epsilon$ -greedy dispatching, and at 1024 each run receives its own 1024 samples for the reason given in Section 6.3. Every entry is the mean over independent runs of that method with the per-instance best of those runs in brackets, the convention of Díaz et al. (2023b)—but the runs are not the same evidence: 30 evolutions for GP and 30 executions for GA against *three* training seeds for the policy, so the two outer brackets are much the stronger selections. Being a best-per-instance, no single run attains a bracketed mean: the best individual rule of the 30 averages 15.3%, 12.8% and 11.0%, against the 12.0%, 10.4% and 8.4% of the bracket. G&T is the hand-crafted baseline, the strongest of Section 5.5. GA’s pair is quoted from Table 2 of Díaz et al. (2023b); the rule the companion study features averages 17.9%, 14.2% and 11.3% and is the value used in Figure 4, Table 9 and the appendix, where the 30 rules were not available. Reference bounds are the best-known expected-makespan values, a different scale from Taillard’s crisp bounds.

| Inst. | rule<br>1 pass | learned<br>one pass |             | learned<br>64 samples |             | learned<br>1024 samples |             | search<br>30 runs |
|-------|----------------|---------------------|-------------|-----------------------|-------------|-------------------------|-------------|-------------------|
|       | G&T            | GP                  | Pol.        | GP                    | Pol.        | GP                      | Pol.        | GA                |
|       |                | mn [bst]            | mn [bst]    | mn [bst]              | mn [bst]    | mn [bst]                | mn [bst]    | mn [bst]          |
| FT10  | 32.2           | 17.0 [7.2]          | 16.7 [16.0] | 11.8 [5.2]            | 8.5 [6.3]   | 8.5 [4.2]               | 6.1 [5.4]   | 5.2 [1.8]         |
| FT20  | 40.0           | 18.5 [9.4]          | 8.6 [4.4]   | 13.3 [5.8]            | 5.8 [4.4]   | 10.3 [5.4]              | 4.4 [3.6]   | 4.4 [1.5]         |
| La21  | 24.1           | 15.9 [10.3]         | 18.3 [15.5] | 12.6 [10.0]           | 12.6 [10.4] | 10.7 [6.8]              | 10.8 [10.7] | 5.0 [3.2]         |
| La24  | 26.3           | 16.1 [10.1]         | 19.3 [17.1] | 11.5 [8.6]            | 11.9 [10.9] | 9.2 [6.2]               | 9.1 [8.1]   | 6.3 [4.1]         |
| La25  | 16.9           | 17.6 [9.6]          | 17.3 [16.7] | 12.3 [8.7]            | 9.6 [9.2]   | 9.3 [5.1]               | 8.9 [8.2]   | 5.1 [1.9]         |
| La27  | 34.8           | 18.3 [12.3]         | 15.0 [13.8] | 12.5 [10.7]           | 11.3 [10.9] | 11.3 [8.2]              | 9.0 [8.1]   | 10.2 [4.6]        |
| La29  | 24.5           | 19.9 [12.9]         | 21.1 [16.2] | 15.0 [11.4]           | 16.6 [14.0] | 12.6 [9.7]              | 13.3 [12.9] | 14.2 [11.1]       |
| La38  | 27.0           | 17.4 [9.2]          | 17.2 [16.6] | 14.3 [9.2]            | 12.8 [11.7] | 12.0 [7.1]              | 9.6 [8.3]   | 9.2 [6.0]         |
| La40  | 27.9           | 16.7 [11.7]         | 12.0 [9.3]  | 12.3 [8.8]            | 9.3 [8.1]   | 10.0 [7.1]              | 7.2 [6.5]   | 8.7 [5.1]         |
| ABZ7  | 28.7           | 17.6 [13.3]         | 16.7 [15.9] | 14.3 [11.7]           | 13.4 [12.0] | 12.2 [8.9]              | 11.9 [11.1] | 12.5 [6.3]        |
| ABZ8  | 36.9           | 24.6 [18.1]         | 21.8 [19.1] | 20.9 [17.0]           | 17.3 [16.9] | 18.6 [15.0]             | 15.8 [15.6] | 18.5 [11.3]       |
| ABZ9  | 41.8           | 29.0 [19.5]         | 22.6 [19.7] | 24.0 [17.9]           | 19.2 [18.8] | 21.5 [17.4]             | 17.1 [16.6] | 18.0 [13.0]       |
| Mean  | 30.1           | 19.0 [12.0]         | 17.2 [15.0] | 14.6 [10.4]           | 12.4 [11.1] | 12.2 [8.4]              | 10.3 [9.6]  | 9.8 [5.8]         |

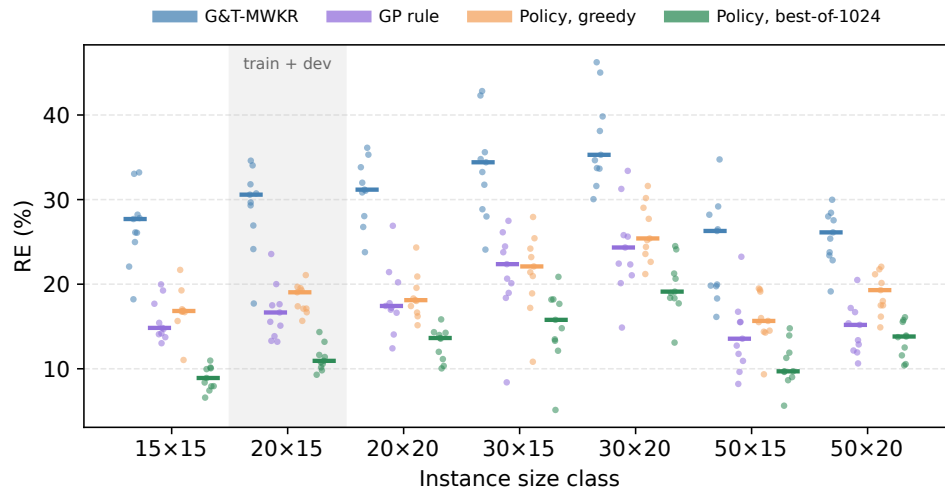
as inference budgets matched, is the subject of dedicated ongoing work. The policy’s standalone RE (13.8% in-size; 10–26% zero-shot) thus sits between dispatching rules (42.3% for the best of them, EST) and per-instance metaheuristics (9.4% mean RE for fEABC over 30 runs)—establishing, for the IJSP, the computational-class trade-off documented for the deterministic JSP (Zhang et al., 2020).



**Fig. 4** The DRL policy against the evolved rule of the companion study (Díaz Rodríguez, 2026b) at matched inference budgets, one point per instance over the 70 interval Taillard instances. Points below the diagonal are instances where the policy wins. Left: one constructive pass each. Right: 1024 samples each, the rule randomized with  $\epsilon$ -greedy dispatching.

**Table 9** Mean RE (%) per size class over the 70 interval Taillard instances. Constructive methods are a single deterministic pass unless stated; the GP rule is the featured evolved rule of the companion study (Díaz Rodríguez, 2026b), also one pass—and the best of that study’s 30 evolutions, selected on these same instances, where the policy rows are means over three seeds (Section 6.4); fEABC is a 30-run average on dedicated hardware, quoted as a reference from a different computational class. The  $20 \times 15$  column ( $\dagger$ ) is the training and development class: it contains the four instances the policy trained on and the six on which the design decisions were made, and is therefore not a generalization result; the other six classes are untouched.

| Method                | $15 \times 15$ | $20 \times 15^\dagger$ | $20 \times 20$ | $30 \times 15$ | $30 \times 20$ | $50 \times 15$ | $50 \times 20$ | All         |
|-----------------------|----------------|------------------------|----------------|----------------|----------------|----------------|----------------|-------------|
| EST                   | 32.3           | 49.2                   | 41.5           | 42.4           | 56.6           | 33.2           | 41.0           | 42.3        |
| MOR                   | 41.4           | 46.6                   | 47.2           | 44.1           | 58.8           | 34.1           | 45.9           | 45.5        |
| G&T-MWKR              | 26.8           | 28.9                   | 30.9           | 33.6           | 36.8           | 23.9           | 25.5           | 29.5        |
| GP rule               | 15.7           | 16.6                   | 18.1           | 21.1           | 24.1           | 13.8           | 14.6           | 17.7        |
| Policy (greedy)       | 16.9           | 18.3                   | 18.5           | 21.2           | 26.1           | 15.8           | 18.8           | 19.4        |
| Policy (best-of-1024) | <b>8.8</b>     | <b>11.2</b>            | <b>12.9</b>    | <b>15.0</b>    | <b>19.6</b>    | <b>10.4</b>    | <b>13.4</b>    | <b>13.0</b> |
| fEABC (30 runs)       | 5.9            | 8.7                    | 10.3           | 9.8            | 16.4           | 5.7            | 9.0            | 9.4         |



**Fig. 5** Per-instance RE by size class over the 70 interval Taillard instances: every instance is a point, bars mark class medians. The  $20 \times 15$  class is shaded because it holds the four training and six development instances; the other six classes were touched by neither.

## 7 Analysis

The results above establish what the policy achieves. This section asks what is doing the work: which inputs the network actually consults, which of the design choices behind it earn their cost, and how dependable the schedules prove when the durations finally materialize.

### 7.1 What the policy attends to

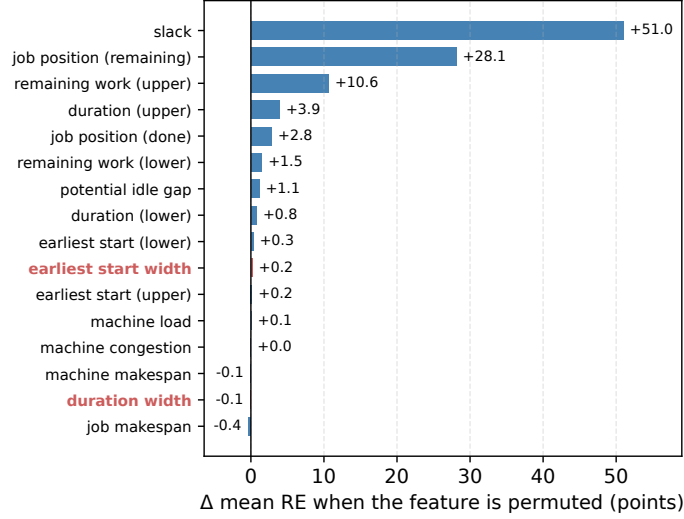
The policy is not interpretable the way a priority expression is, but its inputs can be audited. Figure 6 reports a permutation test: at every decision of a greedy rollout, one feature column is shuffled across the eligible operations, and the resulting mean RE over the development set and the three seeds is compared with the intact rollouts (18.2%).

Three features carry the policy: the relative slack (whose permutation adds 51.0 points of RE), the number of remaining operations in the job (+28.1), and the job’s worst-case remaining work (+10.6)—the attribute families behind MOR and MWKR, and the very attributes the companion study’s evolved expressions select (Díaz Rodríguez, 2026b). The interval-width features contribute nothing measurable at inference (−0.1 to +0.2 points): whatever the policy does with uncertainty, its decisions rest on worst-case magnitudes rather than on interval widths per se.

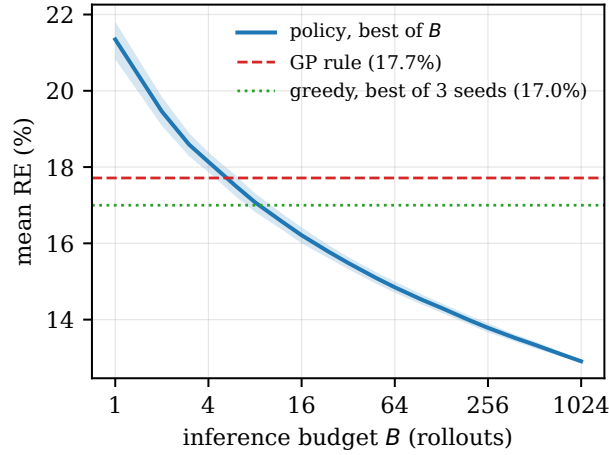
### 7.2 Ablation: best-of- $N$ inference

Introducing best-of-64 inference (over one greedy rollout) improved the full-benchmark mean makespan by 3.6%, strictly better on 3 of 6 development instances and worse on none, while sharply reducing seed variance. Diagnostically, before this change the best training solutions frequently arose in early, exploratory episodes and were never surpassed by the greedy policy—evidence that the policy’s value concentrates in its sampling distribution.

The paper reports the policy at two budgets, one greedy pass and 1024 samples, which leaves the shape between them unstated. To recover it we stored every individual rollout of 342 per checkpoint on each of the 70 instances—71,820 schedules—so that best-of- $B$  can be reconstructed for any  $B$  without evaluating anything again. Figure 7 plots the result. Three readings are worth naming. A single *sampled* rollout is much worse than a greedy one, 21.4% against 17.0%: the argmax is a good decision sequence and the distribution around it is wide. The curve overtakes the evolved rule at  $B=6$  and the greedy pass itself at  $B=12$ , so the budget at which sampling starts paying is small, far below the 64 used throughout this paper. And the returns are close to logarithmic over three orders of magnitude, 14.9% at 64 and 12.9% at 1023, with no knee: within the range tested there is no budget beyond which further sampling stops helping, which is what makes the policy a natural front end for a search procedure that can afford more.



**Fig. 6** Permutation importance of the 16 operation features: change in mean development-set RE when the feature’s column is shuffled across the eligible operations at every decision (greedy rollouts, three seeds; intact baseline 18.2%). The two interval-width features are highlighted.



**Fig. 7** Mean RE over the 70 instances as a function of the inference budget  $B$ , reconstructed from stored rollouts. The band is the 10–90 percentile over resamplings, that is, the spread a practitioner would see across repetitions at that budget, not a confidence interval on the mean. The budget is split across the three checkpoints as the deployed protocol splits it, and the greedy reference is aggregated the same way; the 19.4% of Section 6.4 is the mean over seeds instead. EST, at 42.3%, is off the top of the frame.

**Table 10** Attention ablation: mean [best] RE (%) on the development set.

| Configuration         | 300 eps            | 1000 eps           |
|-----------------------|--------------------|--------------------|
| Deep Sets             | <b>16.6</b> [15.6] | <b>13.4</b> [12.3] |
| + attention ( $B=2$ ) | 17.6 [16.3]        | 15.2 [14.0]        |

### 7.3 Ablation: attention versus pooling

The two-block attention variant of Section 4.4 was evaluated at both operating points with paired seeds: +0.9% mean makespan at 300 episodes (all six instances within the noise threshold) and +1.5% at 1000 (one instance significantly worse), for 3.2 times the parameters and  $\approx 30\%$  higher per-episode cost (Table 10). At these budgets the pooled context already conveys the interaction information the policy can exploit; richer pairwise structure may require larger budgets or instance distributions to pay off.

### 7.4 Ablation: specialist versus multi-size training

Size invariance enables training on mixed size classes, a natural hypothesis for improving transfer. We trained agents on twelve instances spanning  $15 \times 15$ ,  $20 \times 15$  and  $20 \times 20$ , at matched total budget (3000 episodes) and at equalized *per-instance* budget (12,000 episodes—three times the specialist’s total cost), against the  $20 \times 15$  specialist, all sides with best-of-64 inference and averaged over their three seeds. At matched total budget the specialist wins on all five common evaluation instances (10.5 vs. 15.9 on  $15 \times 15$ ; 15.0 vs. 20.0 on  $20 \times 20$ ; 25.9 vs. 30.6 on the unseen  $30 \times 20$ ). Tripling the multi-size budget recovers two of the five (9.6 vs. 10.5 on  $15 \times 15$ ; 15.8 vs. 16.8 on the unseen  $50 \times 20$ ) but not the rest: the specialist stays ahead on  $20 \times 20$  (15.0 vs. 19.4), a class the multi-size agent itself trained on, and on both  $30 \times 15$  and  $30 \times 20$ . At matched cost, per-instance training depth dominates size diversity; even at triple cost, diversity buys back a minority of instances.

Earlier development iterations provide the complementary caution, and two are worth naming. Appending four uncertainty features that are linear combinations of ones already present—the widths and midpoints of quantities the encoder already receives as bounds—cost 9.1% in mean makespan. Normalizing the temporal features by the instance lower bound, an input scaling retrofitted onto an optimizer co-adapted to the raw magnitudes, cost 112.7%: learning collapsed outright. Both figures are against the configuration in force when each was tried, at a reduced screening budget, and both modifications were rejected on that evidence. Representation changes have to be co-designed with the optimization rather than bolted onto it.

### 7.5 What interval awareness contributes

Before reporting these ablations we should be clear about which of them could have come out otherwise, because one of them could not. Componentwise, the upper bound



of a completion obeys

$$c_o^U = \max(c_{JP(o)}^U, c_{MP(o)}^U) + p_o^U, \quad (7)$$

a recursion in upper bounds alone:  $C_{\max}^U$  is a deterministic function of the processing order and the upper endpoints, and the lower endpoints do not appear in it. The reward of Section 4.1 minimizes exactly that quantity—across its six components not one reads a lower bound—so there is no channel through which knowing an interval’s width could reduce what the policy is being trained to reduce. A width feature can at best be an auxiliary signal for learning; it cannot be information about the objective, because the objective does not contain it.

That is worth stating plainly, since it turns the first ablation below from a discovery into a confirmation, and locates the informative experiment elsewhere. The remaining questions are whether the widths help *learning* even though they cannot help the objective (first), whether the upper bounds seen during training must be the true ones (second, which Eq. (7) does not settle, since training on midpoints changes the upper bounds themselves), and what happens when the objective is changed so that width does enter it (third).

The representation of Section 4.2 was designed to make uncertainty visible: interval widths enter as relative magnitudes, and the durations, earliest starts and remaining work of a candidate appear as bounds rather than as point estimates. The permutation audit of Section 7.1 already found the two width features unused at inference. Three training-time ablations ask the stronger questions: whether the policy would be any worse had it never been able to see the uncertainty at all, whether it needs the true bounds during training, and whether it would use the widths if the reward paid for them.

The first removes the information. The encoder collapses every interval to its worst case before the features are computed, so the width features are identically zero and the bounds coincide; the architecture, the hyperparameters and the schedule builder are untouched, and the environment still executes interval semantics. Three seeds retrained from scratch reach 13.62% mean RE on the development set against the 13.44% of the full representation—a difference of 0.18 points, worse on twelve of the eighteen instance–seed pairs and better on six, and not significant (Wilcoxon signed-rank,  $p = 0.47$ ). By Eq. (7) this is the expected outcome, and the experiment is worth running only because the prediction concerns the objective while the measurement concerns learning: the widths could in principle have acted as a useful auxiliary signal, shaping the value function or the exploration even while contributing nothing to the quantity minimized. They do not. The effect size makes the bound precise: the paired differences have a standard deviation of 1.54 points, so 0.18 is 0.12 of one, and detecting an effect that small at conventional power would take on the order of ninety seeds. Not a call for ninety seeds—an observation that whatever the widths contribute is an order of magnitude below the run-to-run noise of training itself.

The second removes the uncertainty from training altogether. Because the benchmark perturbs each crisp duration symmetrically, the original crisp instances *are* the

midpoint scenario of their interval counterparts, which makes them available as a control: we train on them and evaluate, unchanged, on the interval instances. This one is not covered by Eq. (7)—training on midpoints replaces the upper endpoints themselves, so the policy is fitted to a different problem—and it is the ablation where the extra runs changed the answer. Over three seeds the gap was 0.74 points and not significant; over ten it is 0.63 points, from 13.82% to 14.45%, worse on 36 of the 60 instance–seed pairs and positive for eight of the ten seeds ( $p = 0.045$ ). Training on the true upper bounds is therefore slightly better than training on their midpoints. The margin is small—under five per cent of the RE it improves, at a power of about 0.7, so it should be read as a detected effect and not a measured one—but it is there, and the earlier three-seed version of this paragraph claimed its absence.

Taken together with the permutation audit, the delimitation is sharper than a blanket statement about interval awareness. As *input features* the widths are inert: unused at inference, free to remove, and Eq. (7) says why in a way that generalizes past this network—any method minimizing the worst-case makespan of a semiactive schedule optimizes a function of the upper endpoints alone, so interval-aware inputs cannot pay for themselves under that objective, whatever the model class. What is *not* inert is the training distribution: the policy fitted to the real upper bounds beats the one fitted to midpoints, which is exactly the case the recursion leaves open. The boundary of the claim is therefore the reward and the data, not the architecture. Under an objective that does contain the width—which has no counterpart in the crisp problem—the companion study finds its width terminals do earn their place (Díaz Rodríguez, 2026b).

The third ablation therefore moves the boundary itself. We retrain the policy, unchanged in every other respect, under

$$f_\lambda = C_{\max}^U + \lambda (C_{\max}^U - C_{\max}^L), \quad \lambda = 1, \quad (8)$$

which charges the schedule for the width of the interval it predicts and is the analogue of the robust objective of the companion study. Because optimizing one quantity and selecting by another would be incoherent, the best-of- $N$  inference of this arm ranks its samples by  $f_\lambda$  as well. Deployed as that package, the objective does what it promises: over the same eighteen instance–seed pairs the interval narrows from 12.85% to 11.90% of the expected makespan (narrower on 13 of 18,  $p = 0.010$ ) and the expected makespan itself rises by 1.94 points, to 15.38% ( $p = 0.007$ ). Unlike the two ablations above, both effects are significant. A width-rewarding objective is available, and it costs about two points of RE.

The interesting part is where the narrowing comes from. Training and selection changed together above, so we separate them: from a fresh pool of 64 rollouts per instance and seed we reconstruct the best-of-64 of *both* arms under one common criterion, so that only the weights differ. Under that control the two policies become indistinguishable—ranking by the upper bound, the widths are 12.80% for the default arm and 12.69% for the robust one ( $p = 0.93$ ); ranking by  $f_\lambda$ , 11.74% and 12.20% ( $p = 0.12$ ), if anything the wrong way round. What narrows the interval is which sample is kept, not which weights produced it: applying the robust criterion to the *default* policy’s own pool already buys most of the reduction, at a comparable cost in

RE. The network, in other words, is not merely indifferent to the widths in its input; it does not become sensitive to them when the reward asks it to.

The companion study runs the same objective at the same  $\lambda$  and reaches the opposite finding, which makes the pair worth reading together. There, penalizing width narrows the schedules only when the rule has width terminals to act through: sweeping  $\lambda$  from 0 to 4 moves the relative width by 1.64 points for rules that have them and by 0.13—nothing—for rules that do not, from which that study concludes that the objective works *through* the representation (Díaz Rodríguez, 2026b). Our network has both the terminals’ equivalent and the objective, and still its weights do not move; what moves is the ranking among its samples. The same reward therefore reaches the decision function of an evolved rule and does not reach the decision function of a trained policy, at least at this training. The trade itself is priced comparably in the two—the companion pays about 2.5 points of RE per point of width across its sweep and this arm pays 2.0—so the paradigms differ less in what predictability costs than in which part of the system pays it.

That completes the delimitation at a fourth level, and it leaves Section 8 a sharper question than the one it would otherwise inherit: whether any training signal reaches these features, or whether the constructive setting itself decides the width before the policy has a say.

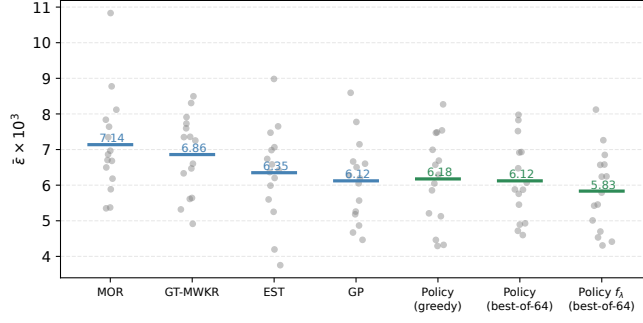
## 7.6 Executorial robustness

A schedule is eventually executed under one realization of the durations, and its interval prediction is useful insofar as the executed makespan stays close to the reported expected value. We adopt the field’s  $\bar{\varepsilon}$  measure (Díaz et al., 2020; Leon, Wu, & Storer, 1994): for a fixed processing order with predicted makespan  $C_{\max}$ ,

$$\bar{\varepsilon} = \frac{1}{K} \sum_{k=1}^K \frac{|C_{\max}^{\text{ex},k} - E[C_{\max}]|}{E[C_{\max}]}, \quad (9)$$

where  $C_{\max}^{\text{ex},k}$  is the crisp makespan of executing the order under the  $k$ -th realization, durations drawn independently and uniformly within their intervals. We use  $K=1000$  common random numbers per instance—identical realizations for every method, drawn from a generator seeded by the instance identifier so that the draws reproduce across runs and machines—and report  $\bar{\varepsilon} \times 10^3$ . The Monte Carlo is the expensive part, so this section uses fifteen instances rather than the full seventy: the six of the development set and one or two per size class (TA1, TA2, TA5, TA21, TA25, TA31, TA41, TA51, TA61), the lowest-numbered of each so that the choice is not a choice. Every execution falls inside the predicted interval, as the monotonicity of Eq. (1) guarantees.

The constructive methods separate into two groups, and the policy is in the better one without leading it (Figure 8). Its schedules deviate by  $\bar{\varepsilon} \times 10^3 = 6.18$  under greedy inference and 6.12 under best-of-64—sampling for a lower worst case neither helps nor harms predictability—against 6.86 for G&T-MWKR and 7.14 for MOR, and those two gaps are systematic (the policy is more faithful on 13/15 and 14/15 instances respectively, paired Wilcoxon signed-rank test,  $p < 0.001$  in both cases). Against the



**Fig. 8** Executional robustness by method: mean  $\bar{\varepsilon} \times 10^3$  (bars) over the fifteen evaluation instances, each shown as a point. Lower is a more faithful a-priori prediction. EST, the evolved rule and the policy are indistinguishable at the top and MOR and G&T-MWKR are not; the robust arm  $f_\lambda$ , the only method here that pays for the interval width, is ahead of all of them but not significantly so.

other two the policy is indistinguishable: EST deviates by 6.35 (9/15,  $p = 0.30$ ) and the evolved rule of the companion study by 6.12 (9/15,  $p = 0.72$ ), level with the policy and well inside the noise.

Four methods therefore share the top of this ranking—a hand-crafted rule, an evolved formula and the policy under two inference budgets—and they have one thing in common: none of them optimizes the width of the interval it predicts. A schedule can deviate little without being short, so faithfulness has to be pursued deliberately rather than inherited from a makespan objective; the companion study reaches the same conclusion from its side, where its rule likewise improves on G&T-MWKR but not on EST until an objective that penalizes width is introduced (Díaz Rodríguez, 2026b).

The robust arm of Section 7.5 is the direct test of that reading, since it is the one method here that does charge for the width, and it comes out on top:  $\bar{\varepsilon} \times 10^3 = 5.83$  with the narrowest predictions of the pool (11.5% of the expected makespan against 12.3% for the default arm). The improvement over the default policy is not significant on fifteen instances ( $-0.29$ , more faithful on 9/15,  $p = 0.23$ ), so this is a direction and not a demonstration. But it is the direction the argument predicts, and it comes at the price Section 7.5 measured: about two points of expected makespan. Whether that trade is worth making is a question about the shop, not about the model—and it is available, which the rest of this section shows it otherwise would not be.

## 7.7 Limitations

Four limitations bound the scope of these conclusions. First, the benchmark follows the symmetric  $\pm 15\%$  generation scheme standard in the IJSP literature; asymmetric intervals, and distributions in which the uncertainty varies sharply from one operation to another, would separate the worst-case and expected-value criteria far more than they do here, and no policy was trained under them. Second, the policy builds semiactive schedules, so the space it searches excludes the active schedules that the conflict-set restriction reaches and the insertion-based ones beyond them; the strongest

constructive baseline of this paper exploits the first of those and the policy still leads it, but the ceiling is real. Third, the ablations are run over three training seeds and compared pairwise on them; ten seeds of the main arm bound how much that choice can matter (Section 6.1), but the ablation arms themselves were not extended, so their differences are stated as paired effects over eighteen instance–seed pairs rather than as claims about the population of seeds. Finally, the comparison against the evolved rule of Section 6.4 matches inference budgets but not training ones, and it compares two published artefacts rather than two paradigms trained under one protocol; that study, which is what Xu et al. (2025) call for, and the use of the policy’s sampling distribution to seed population-based search, are the subject of dedicated ongoing work.

## 8 Conclusions and Future Work

The three questions posed in the introduction now have answers. A constructive policy trained by PPO on four interval instances beats the best hand-crafted constructive baselines by a wide margin on its training class (Q1: 13.8% versus 29.5% for G&T-MWKR and  $\approx 46\%$  for the best plain rule). Because the representation is dimensionless and the encoder permutation-invariant, one network serves the entire benchmark: evaluated zero-shot, it outperforms the best dispatching rule on every tested instance of all seven size classes, and the same weights improve on every rule on all twelve classical instances of the FT, La and ABZ families (Q2). Against a rule evolved by genetic programming the answer is not one-sided but budget-dependent: pass for pass the rule is better, and the policy overtakes it only when both are given 1024 samples (Section 6.4), so what the network contributes over a compact formula is its sampling distribution rather than its argmax. And of the standard toolbox, only best-of- $N$  inference paid off at practical budgets: self-attention did not improve on pooling, multi-size training did not overtake the single-size specialist even at three times its cost, and two independent automatic-configuration campaigns failed to beat the untuned defaults at the operating budget (Q3)—negative results we document with the same care as the positive ones. Three more delimit the interval representation that motivated the design, and they separate two things the design had conflated. As inputs the widths are inert: the policy neither uses them at inference ( $-0.1$  points when they are shuffled) nor suffers when they are removed from training (13.62 against 13.44%,  $p = 0.47$ ), which the closed recursion for the worst-case makespan predicts, since that quantity does not contain a lower bound. As training data the intervals are not inert: fitting the policy to the crisp midpoints instead costs 0.63 points over ten seeds (14.45 against 13.82%,  $p = 0.045$ ). Retraining under an objective that charges for the width closes the case rather than opening it: the predicted interval does narrow (11.90 against 12.85%,  $p = 0.010$ ) at a cost of 1.94 points of RE, but with the selection criterion held fixed the robust weights and the default ones are indistinguishable—what narrows the interval is which sample is kept, not which network produced it. At execution time the policy’s schedules are the most dependable of the constructive pool, matching EST, the robustness benchmark that evolved rules also fail to beat; and a permutation audit shows the network leaning on slack, remaining operations and

worst-case remaining work—the attributes hand-crafted and evolved rules exploit—while ignoring the width features at inference.

Three directions follow. The first is suggested by the ablation of Section 7.4: since a specialist transfers so well and multi-size training buys so little, the productive use of size invariance is probably not to train on many sizes at once but to fine-tune a specialist onto a new class cheaply, a curriculum question this paper leaves open. The second concerns the representation, and the robust arm of Section 7.5 sharpens rather than settles it: paying for the width in the reward narrows the schedules the policy *selects* without measurably changing the policy itself, so the width features remain unconsulted even when the objective would reward consulting them. Whether the gradient simply never reaches them, or whether a semiactive constructive decision determines the width before the policy has any say in it, separates two very different diagnoses and calls for a different experiment than any run here—a reward defined on the width alone would tell them apart. The third is deployment: a policy that prices an instance in milliseconds is a natural generator of initial solutions for population-based search, and the quality of its sampling distribution, documented here at a fixed budget, is what such a use would exploit.

## Appendix A Per-instance results

Table A1 lists, for each of the 70 interval Taillard instances, the crisp lower bound and the RE (%) of the earliest-start rule, of Giffler–Thompson with MWKR tie-breaking, of the evolved rule of the companion study (Díaz Rodríguez, 2026b), and of the policy at its two inference budgets: greedy, averaged over the three training seeds, and best-of-1024. The ten 20×15 instances TA11–TA20 are the training and development set.

## Statements and Declarations

### *Funding.*

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### *Competing interests.*

The author has no competing interests to declare that are relevant to the content of this article.

### *Data availability.*

The benchmark instances are openly available in a Zenodo repository (Díaz Rodríguez, 2026a). Code, benchmark harness, all experiment records (accepted and rejected), final checkpoints and figure-generation scripts will be deposited openly upon submission.

[TODO: DOI del depósito propio al enviar]

**Table A1** Per-instance RE (%) of the constructive methods and of the policy. LB is the crisp Taillard lower bound; Pol. is the greedy policy and Pol.\* the best-of-1024 policy.

| Inst. | LB   | EST  | G&T  | GP   | Pol. | Pol.* | Inst. | LB   | EST  | G&T  | GP   | Pol. | Pol.* |
|-------|------|------|------|------|------|-------|-------|------|------|------|------|------|-------|
| TA1   | 1231 | 49.9 | 26.1 | 19.3 | 16.8 | 11.0  | TA36  | 1819 | 41.1 | 42.8 | 23.8 | 18.9 | 13.3  |
| TA2   | 1244 | 24.9 | 27.8 | 14.2 | 19.3 | 6.6   | TA37  | 1771 | 43.4 | 35.6 | 18.4 | 24.2 | 17.7  |
| TA3   | 1218 | 33.0 | 33.2 | 14.1 | 16.8 | 10.1  | TA38  | 1673 | 45.0 | 28.0 | 20.1 | 23.2 | 14.8  |
| TA4   | 1175 | 32.0 | 27.7 | 13.0 | 21.7 | 10.0  | TA39  | 1795 | 46.6 | 28.9 | 22.4 | 17.2 | 13.5  |
| TA5   | 1224 | 27.2 | 33.0 | 14.7 | 17.0 | 8.4   | TA40  | 1651 | 46.9 | 42.3 | 26.1 | 27.9 | 20.9  |
| TA6   | 1238 | 26.8 | 18.2 | 13.7 | 11.0 | 7.4   | TA41  | 1906 | 44.6 | 45.0 | 33.4 | 31.6 | 24.5  |
| TA7   | 1227 | 30.7 | 26.1 | 17.7 | 16.7 | 7.9   | TA42  | 1884 | 71.1 | 38.1 | 24.3 | 25.4 | 18.3  |
| TA8   | 1217 | 23.7 | 25.0 | 14.8 | 16.8 | 7.9   | TA43  | 1809 | 68.7 | 39.8 | 22.3 | 29.0 | 20.6  |
| TA9   | 1274 | 37.7 | 28.2 | 20.0 | 15.6 | 8.9   | TA44  | 1948 | 52.1 | 34.7 | 25.8 | 22.6 | 19.0  |
| TA10  | 1241 | 37.0 | 22.1 | 15.4 | 16.9 | 10.2  | TA45  | 1997 | 50.2 | 33.7 | 14.9 | 21.2 | 13.1  |
| TA11  | 1357 | 56.2 | 30.7 | 17.5 | 19.5 | 10.6  | TA46  | 1957 | 60.8 | 31.6 | 22.4 | 23.6 | 17.7  |
| TA12  | 1367 | 48.8 | 34.6 | 17.6 | 17.1 | 10.9  | TA47  | 1807 | 46.8 | 30.1 | 25.6 | 25.2 | 19.1  |
| TA13  | 1342 | 60.2 | 26.9 | 13.3 | 19.3 | 13.2  | TA48  | 1912 | 59.1 | 33.7 | 21.1 | 24.4 | 18.4  |
| TA14  | 1345 | 39.5 | 29.7 | 13.2 | 17.4 | 9.3   | TA49  | 1931 | 51.0 | 35.3 | 20.1 | 27.7 | 21.3  |
| TA15  | 1339 | 48.1 | 30.6 | 16.7 | 19.7 | 11.4  | TA50  | 1833 | 61.4 | 46.2 | 31.3 | 30.2 | 24.1  |
| TA16  | 1360 | 45.2 | 34.0 | 15.6 | 17.1 | 10.8  | TA51  | 2760 | 33.8 | 34.7 | 23.2 | 19.1 | 11.9  |
| TA17  | 1462 | 58.9 | 17.7 | 13.9 | 15.7 | 9.8   | TA52  | 2756 | 38.0 | 26.5 | 12.7 | 16.0 | 11.3  |
| TA18  | 1377 | 43.8 | 31.8 | 23.6 | 21.1 | 14.3  | TA53  | 2717 | 34.1 | 18.3 | 11.8 | 15.7 | 9.7   |
| TA19  | 1332 | 45.8 | 29.3 | 20.0 | 16.7 | 11.6  | TA54  | 2839 | 20.9 | 16.1 | 8.2  | 9.3  | 5.6   |
| TA20  | 1348 | 45.7 | 24.1 | 15.1 | 19.0 | 10.2  | TA55  | 2679 | 31.7 | 29.2 | 15.5 | 19.4 | 14.8  |
| TA21  | 1642 | 33.4 | 28.0 | 20.2 | 16.6 | 11.1  | TA56  | 2781 | 34.8 | 19.8 | 10.9 | 14.4 | 8.6   |
| TA22  | 1561 | 44.2 | 30.8 | 17.4 | 20.9 | 14.3  | TA57  | 2943 | 32.9 | 20.0 | 9.6  | 15.5 | 9.6   |
| TA23  | 1518 | 42.8 | 35.3 | 17.8 | 16.2 | 13.6  | TA58  | 2885 | 37.4 | 26.3 | 13.6 | 14.5 | 9.0   |
| TA24  | 1644 | 56.3 | 23.8 | 14.1 | 15.1 | 10.3  | TA59  | 2655 | 26.6 | 28.2 | 16.7 | 19.5 | 13.9  |
| TA25  | 1558 | 45.8 | 33.8 | 16.6 | 19.6 | 14.0  | TA60  | 2723 | 41.7 | 19.8 | 15.5 | 14.3 | 9.7   |
| TA26  | 1591 | 42.8 | 31.2 | 26.9 | 18.0 | 14.3  | TA61  | 2868 | 45.7 | 25.4 | 15.4 | 20.1 | 13.9  |
| TA27  | 1652 | 50.0 | 31.1 | 21.4 | 17.4 | 13.6  | TA62  | 2869 | 43.0 | 27.6 | 17.2 | 21.2 | 15.5  |
| TA28  | 1603 | 36.8 | 26.8 | 12.4 | 18.1 | 10.0  | TA63  | 2755 | 38.8 | 23.8 | 13.4 | 17.5 | 13.7  |
| TA29  | 1583 | 28.8 | 32.0 | 17.2 | 18.3 | 12.0  | TA64  | 2702 | 38.2 | 28.4 | 12.2 | 17.5 | 11.6  |
| TA30  | 1528 | 34.2 | 36.1 | 17.0 | 24.3 | 15.8  | TA65  | 2725 | 42.9 | 30.0 | 20.5 | 22.1 | 16.1  |
| TA31  | 1764 | 30.7 | 34.4 | 24.5 | 21.4 | 12.1  | TA66  | 2845 | 41.8 | 28.0 | 11.9 | 19.3 | 13.8  |
| TA32  | 1774 | 42.7 | 33.3 | 19.0 | 22.1 | 18.2  | TA67  | 2825 | 31.7 | 26.1 | 15.2 | 18.0 | 12.5  |
| TA33  | 1788 | 51.3 | 34.8 | 27.5 | 25.4 | 18.2  | TA68  | 2784 | 50.1 | 19.1 | 10.6 | 14.9 | 10.4  |
| TA34  | 1828 | 45.9 | 31.8 | 20.7 | 21.0 | 15.8  | TA69  | 3071 | 35.3 | 22.8 | 12.9 | 16.2 | 10.6  |
| TA35  | 2007 | 30.3 | 24.1 | 8.4  | 10.8 | 5.1   | TA70  | 2995 | 42.9 | 23.4 | 16.7 | 21.8 | 15.8  |

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